Bridging and Scaling

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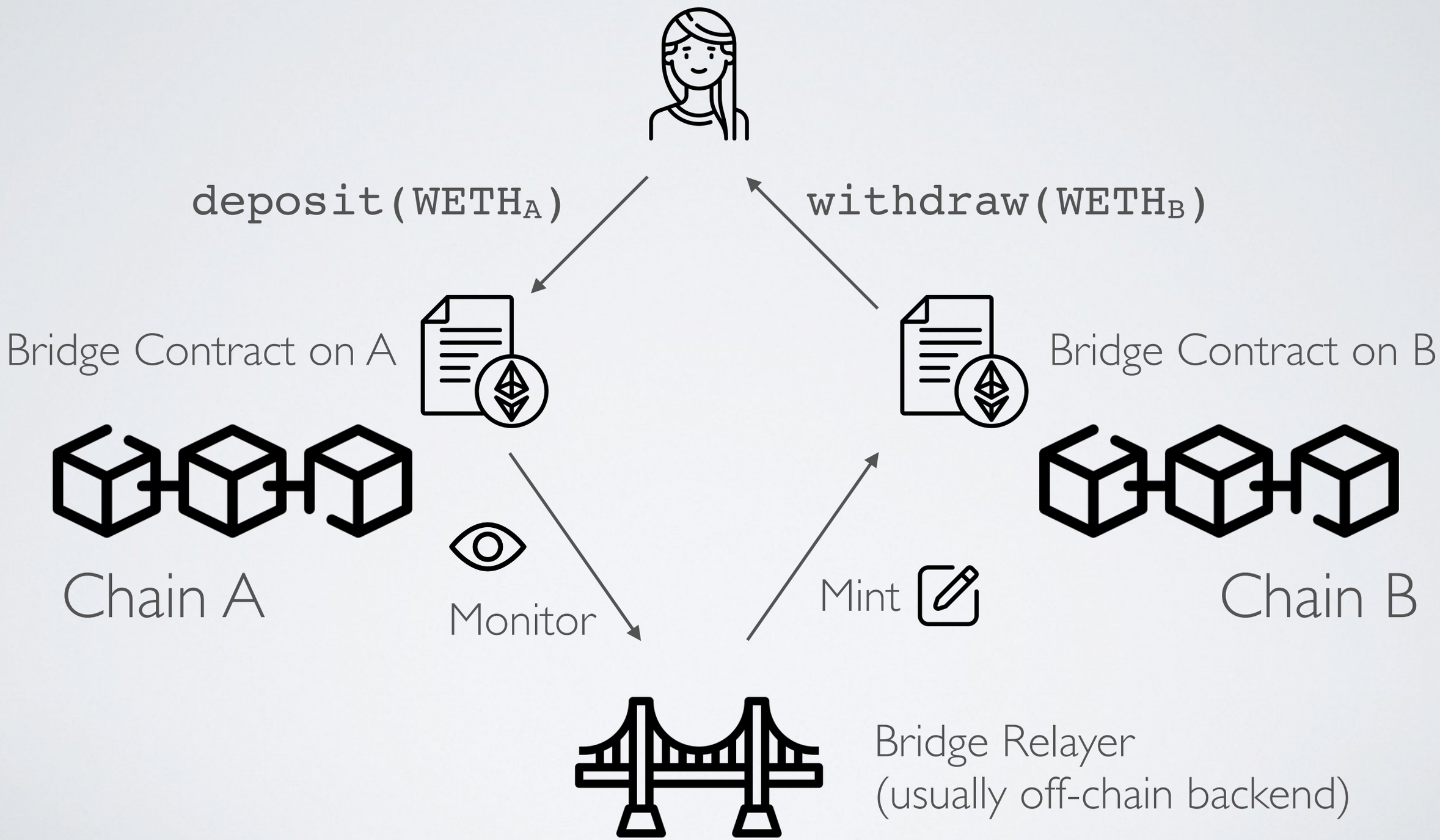
Bridging

Transferring assets from one chain to another (and vice versa)



Scaling

The problem with blockchain mainnets

Transaction Speed

- Bitcoin: 7 tx/s
- Ethereum: 15 tx/s

Transaction Fees

- Bitcoin: 0.000012 BTC (~$1 USD)
- Ethereum: 0.0002 ETH (~ $2 USD)

◉ Transaction speed and transaction fees are **not arbitrary**

➡ They are a direct result of deliberate design choices that **prioritize decentralization** and **security**
i.e the cost to ensure that no single entity can dominate the network

Solutions

- Use a faster consensus
  (hard without compromising with security)
- Split the chains into multiple ones called "shards"
  (work in progress for Ethereum)
- **Rollups** for off-chain or L2-chains

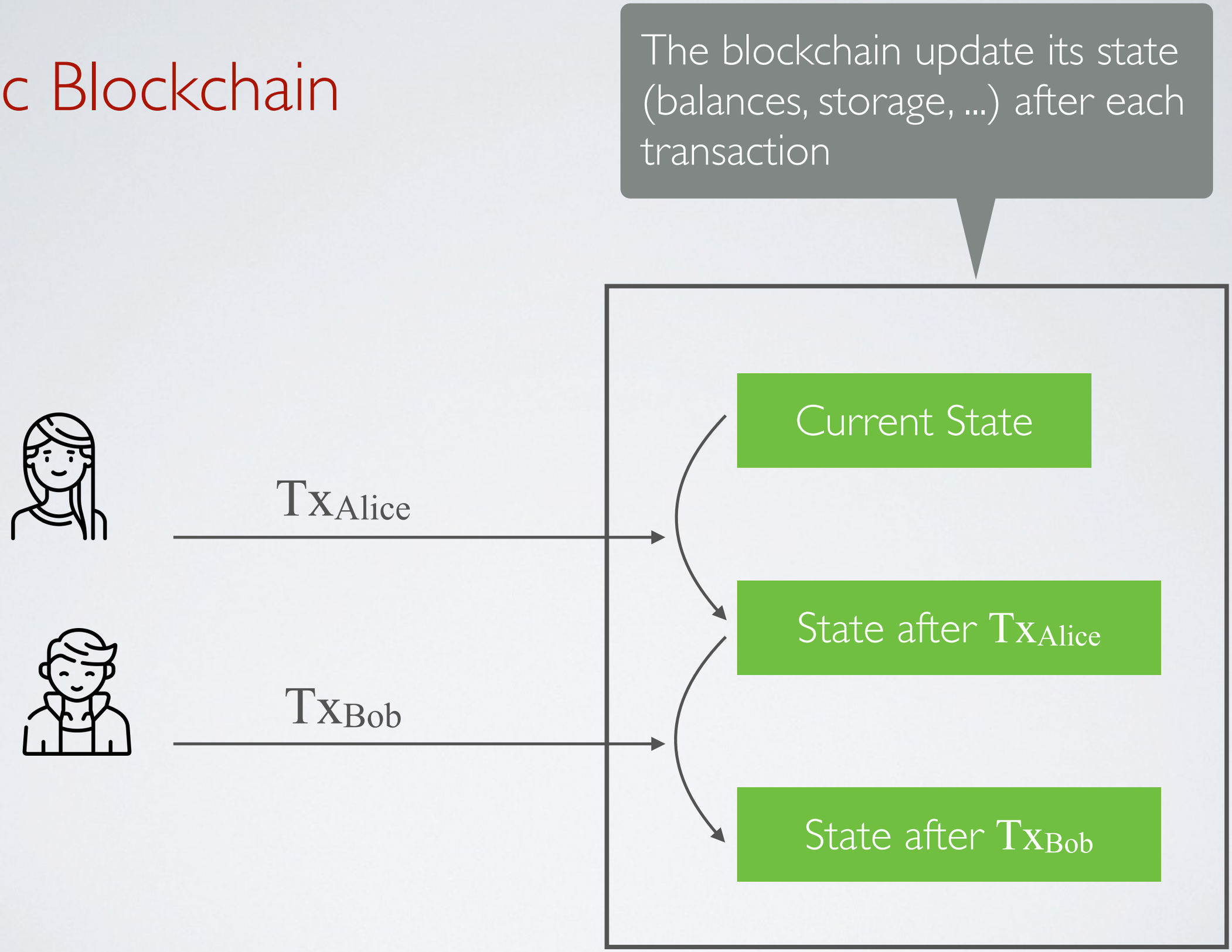
The concept of Rollup

The idea is to process transactions outside of the main chain

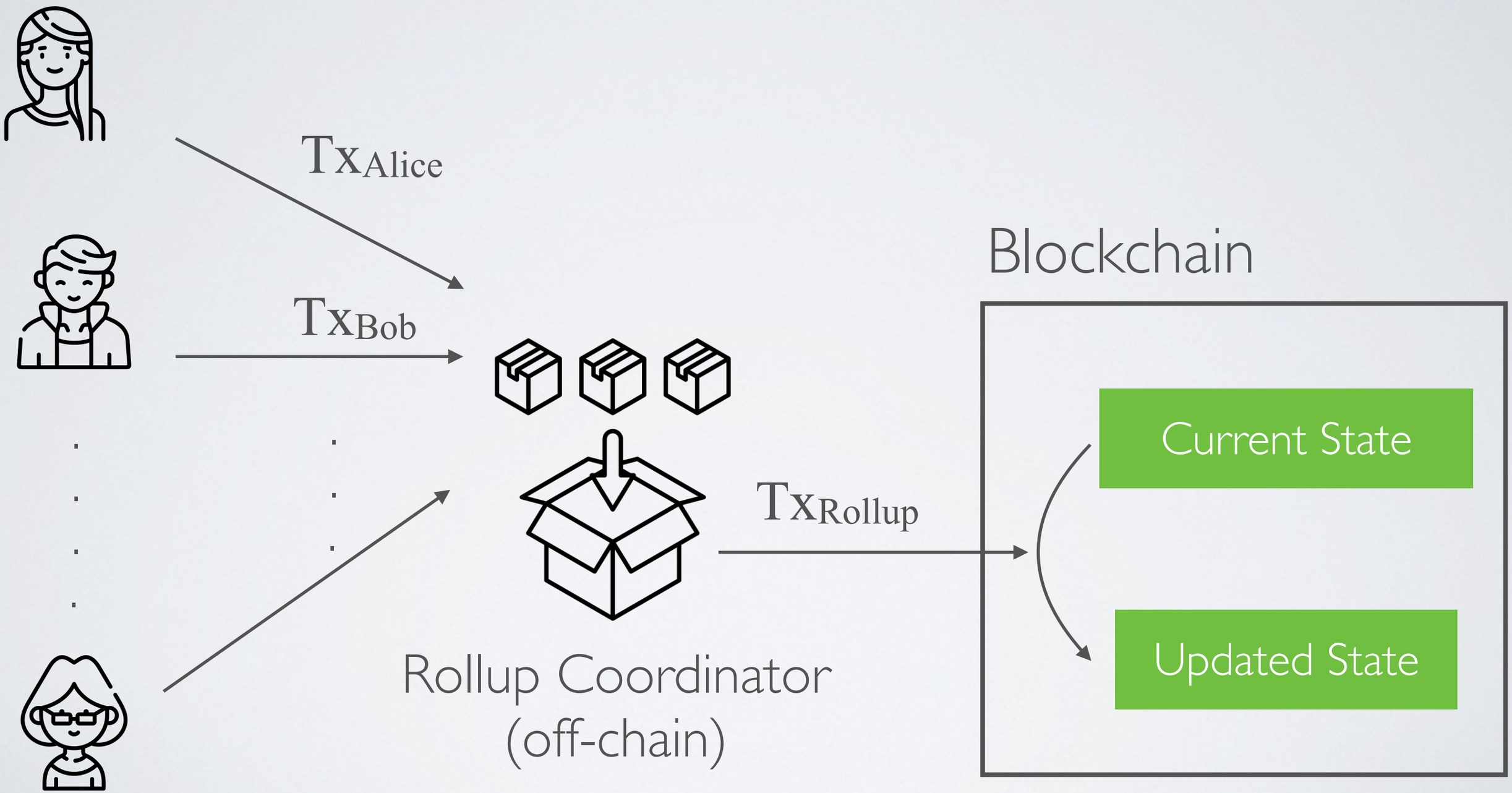
- either off-chain
- or onto another L2-chain (bridging)
  with a faster/cheaper validation

- Both are a necessary trade of with security and decentralization

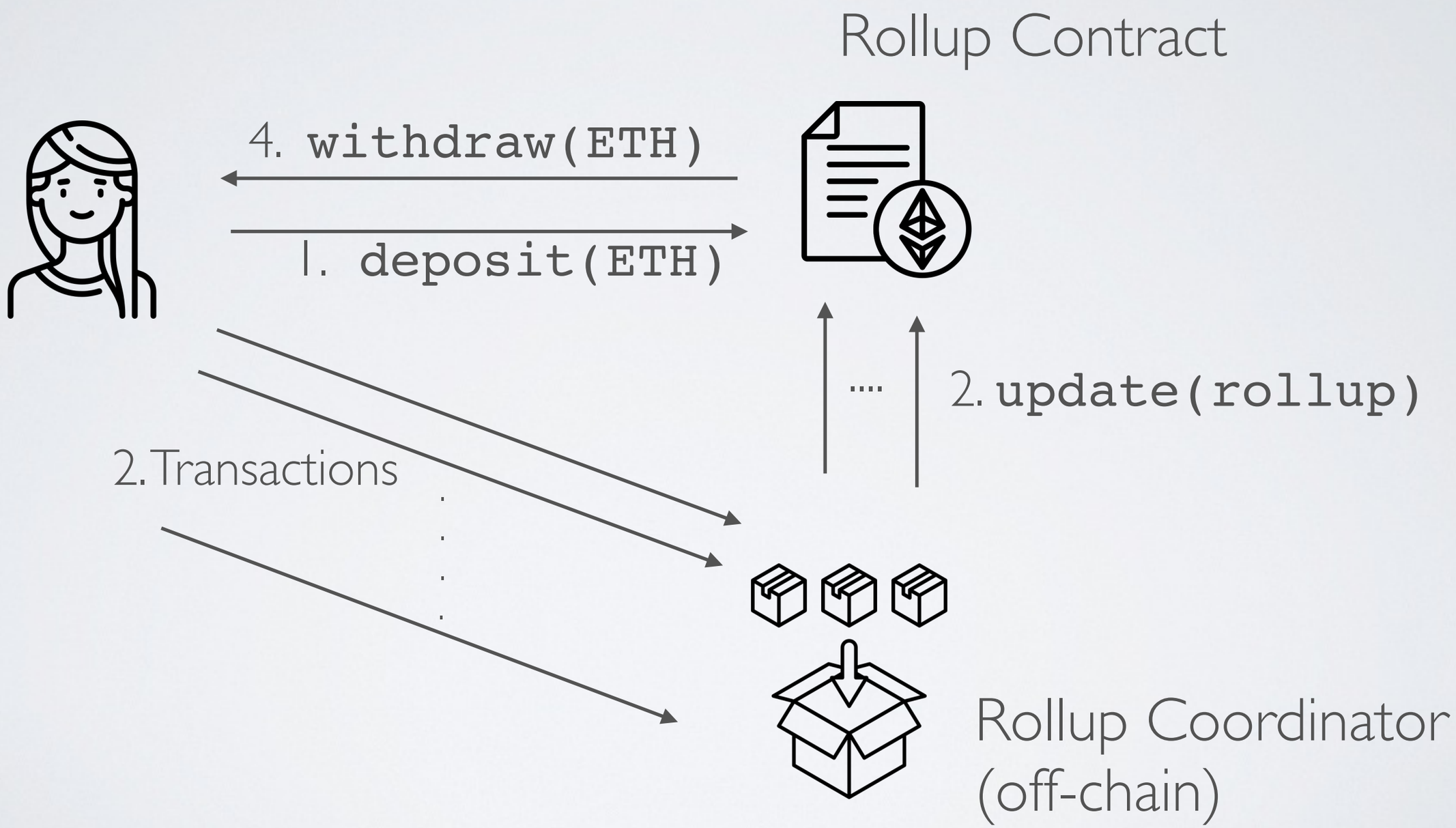
Basic Blockchain



Using a rollup to batch many transactions into one



Using a rollup contract



Three types of Rollup

Naive Rollup (not gas efficient)
The rollup is entirely verified on-chain
(not use in practice because expensive but good to understand the concept of rollup)

Optimistic Rollup (very gas efficient)
The rollup is verified off-chain (after deployment)

Zk-Rollup (pretty gas efficient)
The rollup comes with a ZK-proof that is verified on-chain

Naive Rollup

➡ The rollup contract keeps track of all users' balances

A rollup is the list of transactions verified on-chain by

- verifying each transaction signature
- checking and updating each user's balances accordingly

◉ Not use in practice because gas inefficient

Optimistic Rollup

The rollup contract only stores the root of a Merkle tree with the leafs being the different users balances

➡ When users want to withdraw funds, they need to provide a Merkle proof showing that the pair `(address, balance)` is in the corresponding tree

The rollup is

1. the list of transactions verified off-chained
2. and the new tree root computed off-chain

➡ The rollup coordinator is trusted to compute the right balances and build the right tree

Once the rollup submitted on-chain, verifiers can check it and challenge it during a dispute period (usually 7 days)

- If the rollup is proved fraudulent, it is rolled back and the stake deposited by the rollup coordinator is slashed

Zk-Rollup

The rollup contract stores a Merkle root (similar to Optimistic)

The rollup is

1. the list of transactions is verified off-chained,
2. the new tree root computed off-chain
3. and a ZK-proof proving that given the old tree and the list of transactions, the new tree is correct

✓ The Zk-proof is verified on-chain

Summary on Rollups

Rollups

- Executes transactions off-chain
- Posts transaction data on Ethereum
- Ensures data availability for fraud/validity proofs

- Uses **`calldata`** bytes which are expensive
- Bloats the chain state with data that does not need to be permanent

EIP-4844 Proto-Danksharding

Ethereum blob

- ~128 KB binary data chunk
- Attached to a special transaction type
- Not accessible to the EVM
- Stored temporarily (~18 days) and pruned afterward
- Has separate (and cheaper) blob gas fees

✓ Cheaper rollups

✓ Avoid state bloats

✓ Can be used for other applications